# Course Work Description.

The objective of this assignment is to bridge theoretical machine learning concepts with practical application. You will select a peer-reviewed machine learning research paper (from a pre-approved list or subject to instructor approval), deconstruct its methodology, write a critical academic review, and implement a technical baseline to evaluate the authors' claims.

This is not a traditional summary essay. You are expected to act as a technical reviewer, evaluating the mathematical assumptions, data pipelines, and architectural choices made by the authors.

## Submissions

1. Phase 1: Conceptual Mapping (Deconstruction) - 30 Marks
    a. **Problem Formulation:** Define the core problem using standard machine learning terminology. Identify the specific target variables, input features, and whether the task is supervised, unsupervised, or reinforcement learning.
    b. **Methodological Justification:** Explain *why* the authors chose their specific algorithms or statistical techniques over alternatives.
    c. **Data Pipeline Analysis:** Evaluate the dataset. Details how the authors handle preprocessing, missing values, feature scaling, and class imbalances.
2. Phase 2: Technical Application (Practical Component) - 25 Marks
   Select **one** of the following technical tasks to prove your practical understanding of the paper's domain:
    a) **Option A (Baseline Implementation):** Locate the dataset used in the paper (or a highly similar open-source dataset). Using Python and libraries such as Scikit-Learn or PyTorch, build and train a simple baseline model (e.g., Logistic Regression, Random Forest). Compare your baseline metrics against the authors' advanced methodology.
    b) **Option B (Code Review):** If the authors provided an open-source repository, conduct a rigorous code review. Document any inefficiencies, potential data leakage during cross-validation, lack of random seeds, or architectural flaws.
3. Phase 3: The Technical Review Report – 40 Marks
   Synthesize your findings into a formal review report containing:

a. **Technical Summary:** A concise explanation of the paper's methodology and pipeline.
b. **Methodological Critique:** A critical evaluation of the execution. Did the authors use appropriate evaluation metrics? Are their conclusions genuinely supported by the results, or is there evidence of overfitting/overclaiming?
c. **Proposed Alternatives:** Propose and justify at least one alternative algorithmic approach or statistical model that the authors did not utilize, explaining why it could yield better results or computational efficiency.
d. **Stress-Testing:** Formulate a hypothesis on how the authors' model would perform under different constraints (e.g., if the data dimensionality was significantly increased, or if noise was introduced).

## Deliverables

You must submit the following to the submission portal by the deadline:

1. **The Review Report (PDF):** A 2500 word document structured clearly with headings for the three phases outlined above. Ensure proper academic citations are used throughout.
2. **Technical Appendix (Code):** A Jupyter Notebook (.ipynb) or Python script (.py) containing your baseline implementation or detailed code review annotations. The code must be clean, commented, and reproducible.